

NumPy for AI/ML - Complete Guide with Examples, Input/Output, Use Cases & Interview Questions

1. Introduction to NumPy

NumPy (Numerical Python) is the core library for numerical computing in Python and is the foundation

2. Array Creation Methods

```
np.array([1,2,3])
Input: [1,2,3]
Output: [1 2 3]

np.zeros((2,3))
Output:
[[0. 0. 0.]
 [0. 0. 0.]]

np.ones((2,2))
Output:
[[1. 1.]
 [1. 1.]]

np.empty((2,2))
Output varies (uninitialized memory)

np.full((2,2),7)
Output:
[[7 7]
 [7 7]]

np.eye(3)
Output:
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

3. Array Attributes

```
shape, ndim, size, dtype

arr=np.array([[1,2,3],[4,5,6]])

arr.shape -> (2,3)
arr.ndim -> 2
arr.size -> 6
arr.dtype -> int64
```

4. Indexing & Slicing

```
arr=np.array([10,20,30,40,50])

arr[0] -> 10
arr[-1] -> 50
arr[1:4] -> [20 30 40]
```

5. Reshape & Flatten

```
np.arange(12).reshape(3,4)

flatten()
ravel()
transpose()
```

6. Mathematical Operations

```
Addition: a+b
Subtraction: a-b
Multiplication: a*b
Division: a/b
Power: a**2
sqrt, exp, log
```

7. Statistics

```
np.sum()
np.mean()
np.median()
np.std()
np.var()
np.max()
np.min()
np.percentile()
np.quantile()
```

8. Boolean Masking

```
arr[arr>25]
Used for filtering datasets in ML preprocessing.
```

9. Broadcasting

```
arr=np.array([1,2,3])
arr+10
```

```
Output:
[11 12 13]
```

10. Random Module

```
np.random.rand()
np.random.randn()
np.random.randint()
np.random.choice()
np.random.shuffle()
np.random.seed()
```

11. Sorting & Searching

```
np.sort()
np.argsort()
np.where()
np.argmax()
np.argmin()
np.unique()
```

12. Stacking & Splitting

```
np.concatenate()
np.vstack()
np.hstack()
np.column_stack()
np.split()
```

13. Linear Algebra

```
np.dot()
np.matmul()
```

```
A @ B
np.linalg.inv()
np.linalg.det()
np.linalg.eig()
np.linalg.svd()
```

14. File Handling

```
np.save()
np.load()
np.savetxt()
np.loadtxt()
```

15. AI/ML Use Cases

1. Feature Engineering
2. Data Cleaning
3. Normalization
4. Neural Network Matrix Operations
5. Recommendation Systems
6. Computer Vision Pixel Processing
7. Statistical Analysis
8. Deep Learning Tensor Preparation
9. Time Series Analysis
10. Scientific Computing

16. ML Example

```
import numpy as np

X=np.array([[25,50000],
            [30,60000],
            [35,80000]])

X_norm=(X-np.mean(X,axis=0))/np.std(X,axis=0)

Output:
Normalized feature matrix
```

17. Most Used NumPy Methods

```
array, zeros, ones, empty, full, eye,
arange, linspace, reshape, flatten, ravel,
transpose, mean, median, std, var, sum,
max, min, percentile, unique, sort,
argsort, where, argmax, argmin,
concatenate, stack, split, dot, matmul,
inv, eig, det, save, load
```

18. Top Interview Questions

1. Why is NumPy faster than Python lists?
2. What is vectorization?
3. What is broadcasting?
4. Difference between reshape() and resize()?
5. Difference between flatten() and ravel()?
6. What is ndarray?
7. What is axis=0 and axis=1?
8. Difference between np.dot() and np.matmul()?
9. What is a view vs copy?
10. Why use float32 in ML?
11. Explain Boolean Masking.
12. How does NumPy save memory?
13. What are strides?
14. What is the role of NumPy in Deep Learning?
15. Explain matrix multiplication with example.